Table 1. Data sources

Dataset / endpoint	Cases	Controls	Exposure or clinical role
eQTLGen blood expression	—	—	Exposure: up to 31,684; predominantly European
BBJ GD (GCST90018627)	2,809	172,656	Discovery; East Asian
FinnGen GD (E4_GRAVES_STRICT)	3,962	496,386	Post hoc GD comparison; Finnish
UKB hyperthyroidism (GCST90038636)	3,731	480,867	Self-report; ancestry not established as exclusively European
FinnGen GO (GRAVES_OPHT)	858	499,490	Original ophthalmopathy comparison; Finnish
FinnGen GO strict (E4_GRAVES_OPHT_STRICT)	753	499,595	Post hoc definition sensitivity; Finnish

GD, Graves disease; GO, Graves ophthalmopathy; HT, hyperthyroidism/thyrotoxicosis. The three FinnGen R12 endpoints overlap and use population non-cases for their respective definitions. Strict is the source endpoint label, not proof of perfect specificity. The descriptive orbital series (four TED patients, one control) is separate from the genetic datasets.

Table 2. *TSHR*, *IGF1R* and *CTLA4*

Gene	Dataset	IVs	OR (95% CI)	*P* value	PP.H2	PP.H3	PP.H4
TSHR	BBJ GD	1	0.12 (0.07–0.21)	**1.09×10^{-14}**	<0.001	0.049	**0.951**
TSHR	FinnGen GD	1	0.08 (0.05–0.12)	**3.19×10^{-32}**	<0.001	0.736	0.264
TSHR	UKB HT	1	0.09 (0.06–0.14)	**8.77×10^{-28}**	<0.001	0.774	0.226
TSHR	FinnGen GO	1	0.10 (0.04–0.24)	**2.82×10^{-7}**	<0.001	0.014	**0.986**
TSHR	FinnGen GO strict	1	0.15 (0.06–0.39)	**9.02×10^{-5}**	0.027	0.082	**0.891**
IGF1R	BBJ GD	4	1.56 (1.07–2.28)	**0.0212**	0.690	0.236	0.073
IGF1R	FinnGen GD	3	1.07 (0.86–1.33)	0.539	0.731	0.259	0.010
IGF1R	UKB HT	4	1.35 (1.07–1.70)	**0.0117**	0.400	0.196	0.404
IGF1R	FinnGen GO	3	1.41 (0.85–2.33)	0.182	0.623	0.346	0.032
IGF1R	FinnGen GO strict	3	1.52 (0.92–2.49)	0.0996	0.571	0.391	0.038
CTLA4	BBJ GD	1	0.18 (0.11–0.27)	**5.45×10^{-15}**	<0.001	0.799	0.201
CTLA4	FinnGen GD	2	0.17 (0.08–0.36)	**2.15×10^{-6}**	<0.001	0.746	0.254
CTLA4	UKB HT	2	0.21 (0.08–0.57)	**0.0022**	<0.001	0.047	**0.953**
CTLA4	FinnGen GO	2	0.17 (0.04–0.68)	**0.0118**	<0.001	0.022	**0.978**
CTLA4	FinnGen GO strict	2	0.17 (0.05–0.64)	**0.00888**	<0.001	0.021	**0.979**

GD, Graves disease; GO, Graves ophthalmopathy; HT, hyperthyroidism/thyrotoxicosis. IVs, instrumental variables (retained genetic variants). Wald ratio for one IV; random-effects IVW for multiple IVs. ORs are per approximately one SD higher reconstructed blood expression. Bold *P* values denote nominal $P < 0.05$ only. Discovery uses 0.05/2,544. PP.H2 supports an expression-only signal; PP.H3 supports distinct variants; PP.H4 supports one shared variant. Bold PP.H4 denotes ≥0.80 under $p12 = 10^{-5}$. These are probabilities, not *P* values. *CTLA4* multi-IV results are concentrated in rs13030124; *TSHR* is single-IV in all datasets.

Table 3. Screen and supporting analyses

A. BBJ discoveries and FinnGen GD comparison

BBJ discovery gene	BBJ OR (95% CI)	BBJ *P*	FinnGen GD OR (95% CI)	FinnGen GD *P*	Focused support	GD PP.H4
HLA-A	1.98 (1.73–2.27)	2.60×10^{-23}	1.08 (0.77–1.53)	0.648	No	<0.001
HLA-DQA2	2.39 (1.95–2.93)	8.30×10^{-17}	1.90 (1.64–2.20)	1.41×10^{-17}	Yes	<0.001
CTLA4	0.18 (0.11–0.27)	5.45×10^{-15}	0.17 (0.08–0.36)	2.15×10^{-6}	Yes	0.254
TSHR	0.12 (0.07–0.21)	1.09×10^{-14}	0.08 (0.05–0.12)	3.19×10^{-32}	Yes	0.264
C4A	0.45 (0.35–0.57)	2.23×10^{-10}	0.41 (0.26–0.63)	4.97×10^{-5}	Yes	<0.001
HSD3B7	0.29 (0.18–0.47)	2.04×10^{-7}	0.90 (0.72–1.13)	0.372	No	0.011
TUBB	1.49 (1.28–1.74)	3.14×10^{-7}	3.28 (2.09–5.17)	2.68×10^{-7}	Yes	<0.001
VKORC1	0.13 (0.06–0.29)	6.35×10^{-7}	NA	NA	NA	0.011
TNFSF14	0.63 (0.52–0.76)	1.48×10^{-6}	0.77 (0.65–0.92)	0.00361	Yes	0.303
PRSS36	6.00 (2.87–12.57)	2.02×10^{-6}	1.40 (1.00–1.96)	0.0490	No	0.063
MAPKAPK5	0.977 (0.967–0.987)	5.28×10^{-6}	0.990 (0.929–1.055)	0.751	No	0.041
PSMB8	1.25 (1.13–1.38)	6.86×10^{-6}	0.96 (0.85–1.08)	0.512	No	<0.001
IFNGR1	2.10 (1.51–2.91)	9.41×10^{-6}	1.62 (1.18–2.24)	0.00330	Yes	0.313

B. Additional FinnGen GD discoveries

Additional gene	GD IVs	GD OR (95% CI)	GD *P*	BBJ PP.H4	GD PP.H4	GO strict PP.H4
APOM	1	0.05 (0.04–0.06)	1.43×10^{-108}	<0.001	<0.001	0.227
NEU1	1	0.04 (0.03–0.06)	1.43×10^{-108}	<0.001	<0.001	0.061
IGLV2-18	1	0.56 (0.48–0.66)	7.52×10^{-12}	0.029	0.020	0.100
IGLV3-12	1	1.20 (1.14–1.26)	7.52×10^{-12}	0.019	0.022	0.101
TNF	1	0.60 (0.51–0.71)	4.65×10^{-10}	<0.001	<0.001	<0.001
HIST1H4E	1	2.54 (1.88–3.43)	1.51×10^{-9}	<0.001	<0.001	0.007
LTA	1	0.46 (0.35–0.61)	3.75×10^{-8}	<0.001	<0.001	<0.001
LTB	1	0.18 (0.10–0.36)	4.89×10^{-7}	<0.001	<0.001	<0.001
C4B	2	1.63 (1.31–2.03)	1.10×10^{-5}	<0.001	<0.001	0.311

C. Frequency-sensitive results

Gene / GD analysis	Reference OR (95% CI)	Reference *P*	Cohort-frequency OR (95% CI)	Cohort *P*	Interpretation
C4B	1.63 (1.31–2.03)	1.10×10^{-5}	1.75 (1.31–2.33)	1.48×10^{-4}	Heterogeneous; loses screen significance
HLA-G	0.86 (0.72–1.03)	0.107	0.60 (0.47–0.75)	1.72×10^{-5}	Weight-sensitive; single-variant dependence

Panel A: focused support requires concordant direction and $P < 0.05/13$; 7 of 13 qualify. NA is unavailable, not a negative result. Panel B: nine additional reference-frequency FinnGen GD discoveries at $P < 0.05/2{,}544$; post hoc and not independent of the full screen. Seven of the nine are instrumented within the extended MHC region on chromosome 6. *APOM*/*NEU1* share rs3101018; *IGLV2-18*/*IGLV3-12* share rs5759410. The lead *IGLV* expression variant is absent in BBJ, weakening interpretation of its low PP.H4. Panel C: *HLA-G* was added because of frequency sensitivity, not primary discovery. Neither *C4B* nor *HLA-G* gains strong GD colocalization. All 24 genes and five datasets are available in the workbook.

Figure 1

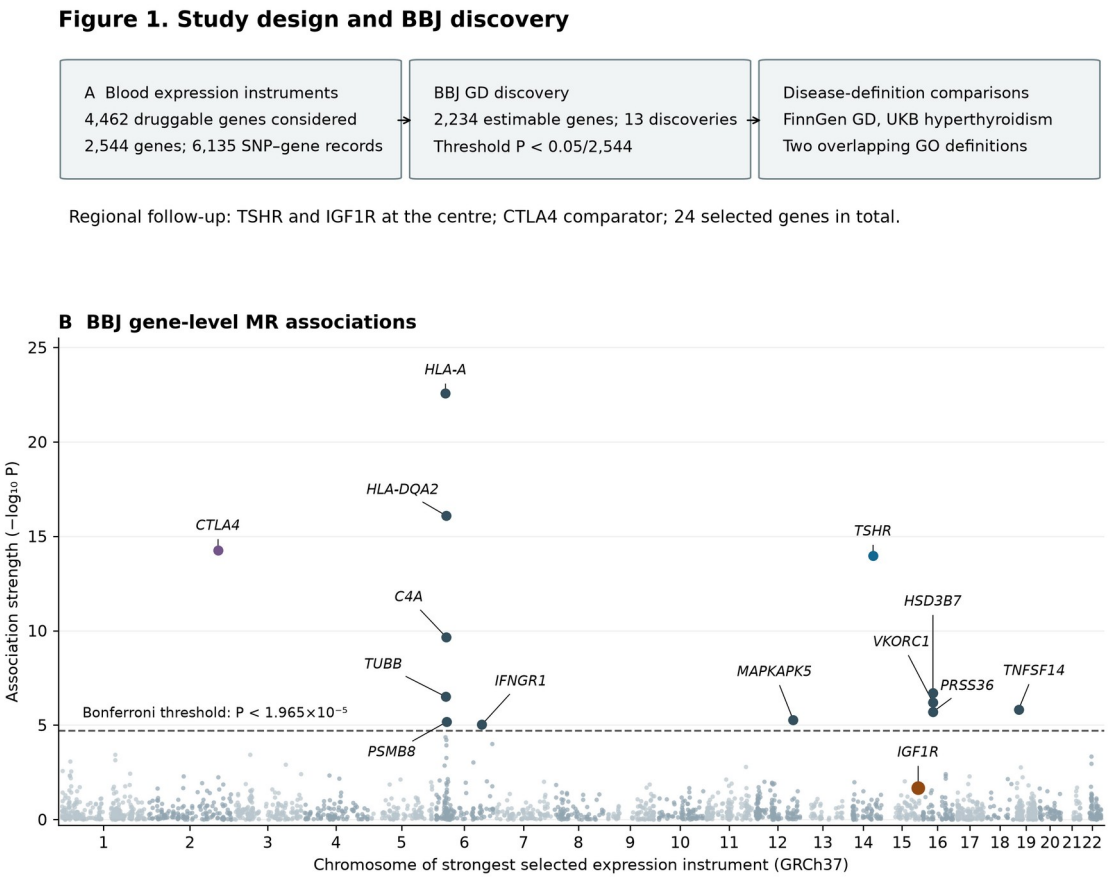


Figure 1. Study design and BBJ discovery. (A) Original blood-expression screen and subsequent disease-definition comparisons. FinnGen endpoints overlap and are not independent replications. (B) One point per estimable gene (n = 2,234), positioned by the GRCh37 coordinate of its strongest selected expression instrument. The horizontal line is $P = 0.05/2{,}544$, equivalent to $-\log10(P) \approx 4.707$. Thirteen discoveries and *IGF1R* are labelled. This is a gene-level MR plot, not a SNP GWAS Manhattan plot. The threshold adjusts the original discovery screen, not every later exploratory analysis.

Figure 2

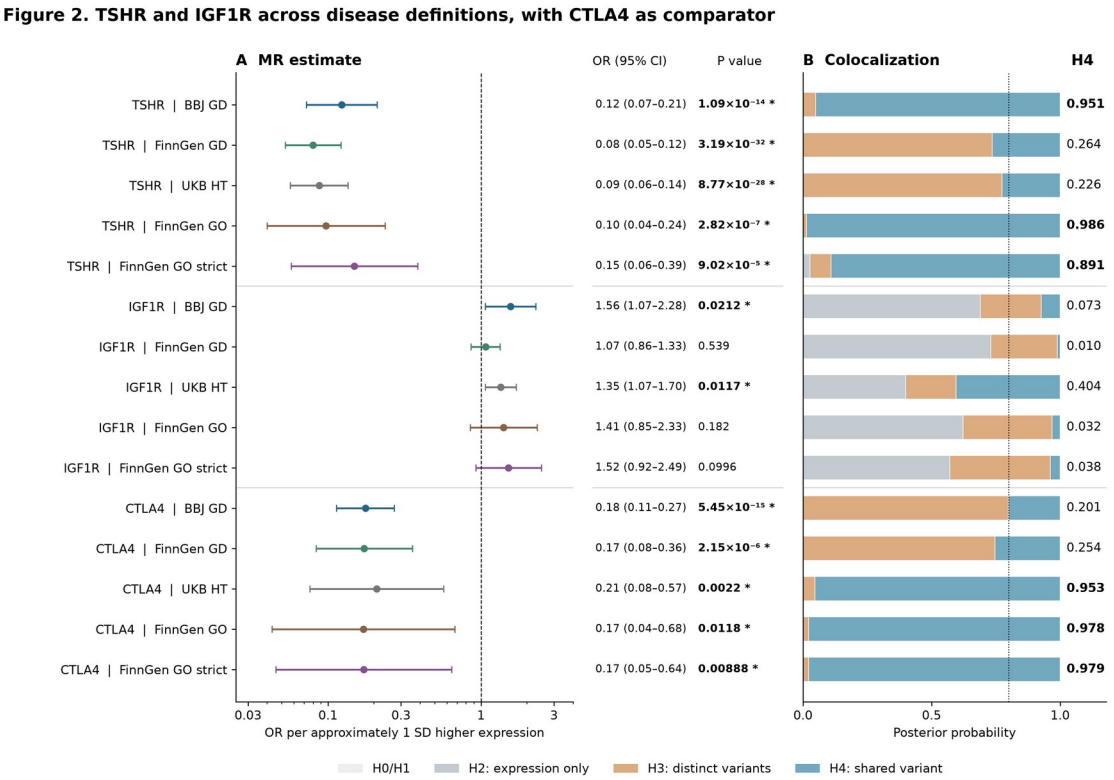


Figure 2. *TSHR* and *IGF1R* across disease definitions, with *CTLA4* as comparator. (A) Wald/IVW odds ratios and 95% CIs for all 15 gene–dataset comparisons. The horizontal axis is logarithmic and the reference is OR = 1. Bold *P* values and an asterisk indicate nominal $P < 0.05$, not Bonferroni significance. ORs describe reconstructed blood expression, not a treatment intervention. (B) Aligned colocalization probabilities under p1 = p2 = 10^{-4} and p12 = 10^{-5}. H0/H1 denotes no association or disease association only; H2 expression only; H3 distinct variants; H4 a shared variant. Bold H4 numbers indicate ≥0.80. MR *P* values and posterior probabilities answer different questions.

Figure 3

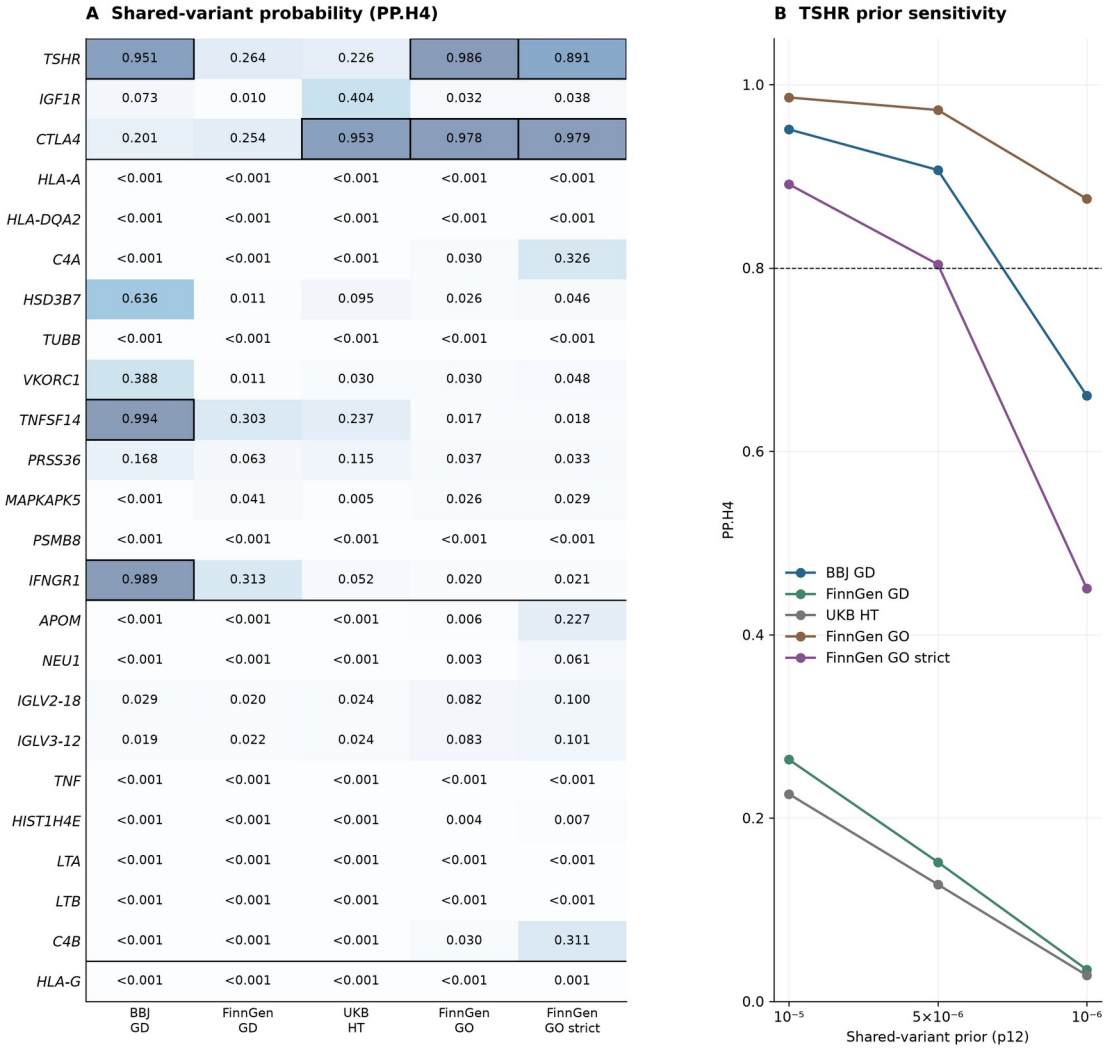


Figure 3. Regional genetic evidence and *TSHR* prior sensitivity. (A) Primary PP.H4 for 24 selected genes and five datasets. Borders mark PP.H4 ≥ 0.80, not *P*-value significance. All lettering is black, and probabilities <0.001 retain their full precision in the workbook. Row dividers separate the three focal/comparator genes, the remaining BBJ discoveries, nine additional FinnGen GD discoveries, and sensitivity-selected *HLA-G*. Only these selected genes underwent colocalization. The low BBJ *IGLV* probabilities are limited by absence of the lead expression variant. (B) *TSHR* PP.H4 at p12 = 10^{-5}, 5×10^{-6} and 10^{-6}; p1 = p2 = 10^{-4}. The dashed line marks the reporting threshold of 0.80. FinnGen GO and GO strict are overlapping ophthalmopathy endpoints; GD is E4_GRAVES_STRICT.

Table S1. *TSHR* regional and prior sensitivity

Dataset	Full cis SNPs	Primary PP.H4	Lower-prior PP.H4	PP.H4 range across windows
BBJ GD	4,336	0.951	0.661	0.951–0.951
FinnGen GD	6,579	0.264	0.035	0.264–0.264
UKB HT	5,667	0.226	0.028	0.226–0.226
FinnGen GO	6,579	0.986	0.875	0.986–0.987
FinnGen GO strict	6,579	0.891	0.451	0.891–0.905

Lower prior: $p12 = 10^{-6}$; $p1 = p2 = 10^{-4}$ throughout. Window range uses the full source cis-region, ±500 kb, ±250 kb and five-outcome intersection at the primary prior. It is not a confidence interval. Full regional disease lead variants were retained. All 240 settings are in TSHR_windows.

Table S2. Frequency-sensitive exploratory candidates

Gene / dataset	Frequency	IVs	OR (95% CI)	*P* value	Q-test *P*
C4B / FinnGen GD	Reference	2	1.63 (1.31–2.03)	1.10×10^{-5}	7.79×10^{-7}
C4B / FinnGen GD	eQTLGen	2	1.75 (1.31–2.33)	1.48×10^{-4}	1.36×10^{-8}
C4B / FinnGen GO strict	Reference	2	1.68 (1.44–1.95)	2.74×10^{-11}	0.122
C4B / FinnGen GO strict	eQTLGen	2	1.80 (1.46–2.23)	6.17×10^{-8}	0.0587
HLA-G / FinnGen GD	Reference	16	0.86 (0.72–1.03)	0.107	9.98×10^{-24}
HLA-G / FinnGen GD	eQTLGen	16	0.60 (0.47–0.75)	1.72×10^{-5}	2.26×10^{-10}
HLA-G / FinnGen GO strict	Reference	16	0.91 (0.76–1.08)	0.267	0.0231
HLA-G / FinnGen GO strict	eQTLGen	16	0.70 (0.53–0.92)	0.0122	0.131

C4B and *HLA-G* frequency sensitivity. Variant sets and outcome effects were held fixed. Frequency changes rescaled expression estimates and IVW weights. Q tests heterogeneity, not directional pleiotropy; its *P* value is not a gene-association *P* value. All 72 omission estimates and instrument weights are provided in the workbook.

Table S3. Tissue-specific expression and FinnGen GD

Gene / tissue	N	IVs	OR (95% CI)	*P* value	Adjusted *P*
TSHR / Thyroid	574	0	No qualifying IV	—	—
IGF1R / Thyroid	574	1	0.76 (0.56–1.02)	0.0692	0.692
TSHR / Cultured fibroblasts	483	0	Not tested	—	—
IGF1R / Cultured fibroblasts	483	1	1.11 (0.89–1.38)	0.359	1.000
TSHR / Subcutaneous adipose	581	1	0.36 (0.31–0.43)	1.28×10^{-34}	1.28×10^{-33}
IGF1R / Subcutaneous adipose	581	0	No qualifying IV	—	—
TSHR / Visceral adipose	469	1	0.50 (0.43–0.58)	4.09×10^{-19}	4.09×10^{-18}
IGF1R / Visceral adipose	469	0	No qualifying IV	—	—
TSHR / Whole blood	670	0	No qualifying IV	—	—
IGF1R / Whole blood	670	0	No qualifying IV	—	—

Fixed-panel tissue extension with FinnGen GD as the primary disease dataset. IVs are retained instrumental variants. ORs are per unit of normalized, inverse-normal-transformed expression. Adjusted $P = \min(1, 10P)$; the ten-comparison denominator includes unestimable cells. Not tested means the gene was absent from the tissue eGenes file; no qualifying IV means the minimum eQTL *P* did not meet 5×10^{-8}. All four estimable combinations used one SNP (F 32.1–61.1), so each *P* value equals that variant's disease-association *P* value. The adipose *TSHR* instruments (rs12101255 and rs74064796, 50 bp apart in *TSHR* intron 1) are themselves blood *TSHR* eQTLs near rs179252 and therefore re-express the regional *TSHR* association rather than providing independent evidence. The 40 secondary comparisons use a separate denominator of 40 and are retained, with all source statistics, in Tissue_extension. Cultured fibroblasts and adipose tissues are not orbital tissue. Original detection benchmarks remain in Detection_original.

Figure S1

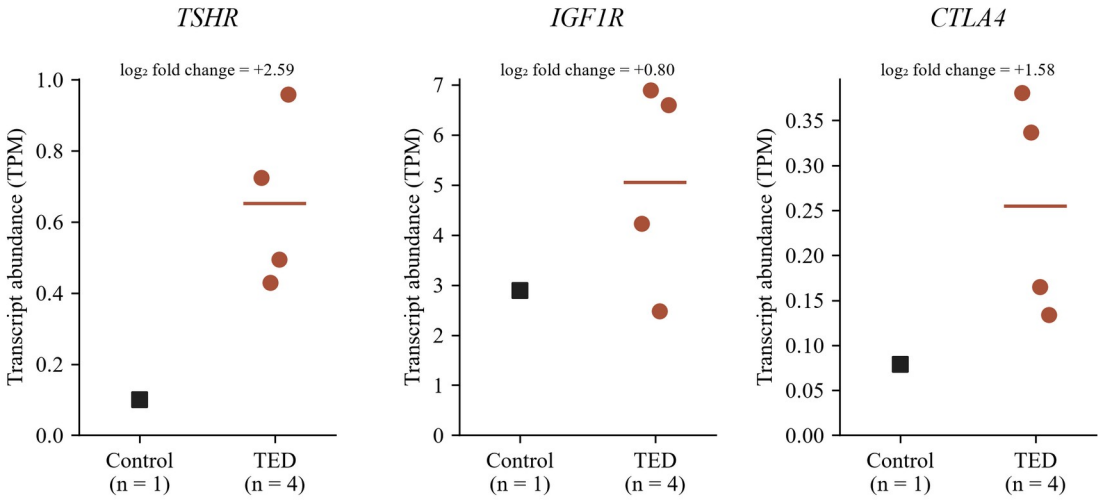


Figure S1. Descriptive orbital expression. Sample-level transcript abundance for *TSHR*, *IGF1R* and *CTLA4* in four TED patients and one control. Points are biological samples; horizontal lines show TED means. Fold changes use log2[(mean TED TPM + 0.01)/(control TPM + 0.01)]. Axes differ by gene. No inferential expression test was performed; these observations do not validate the genetic findings.

Figure S2

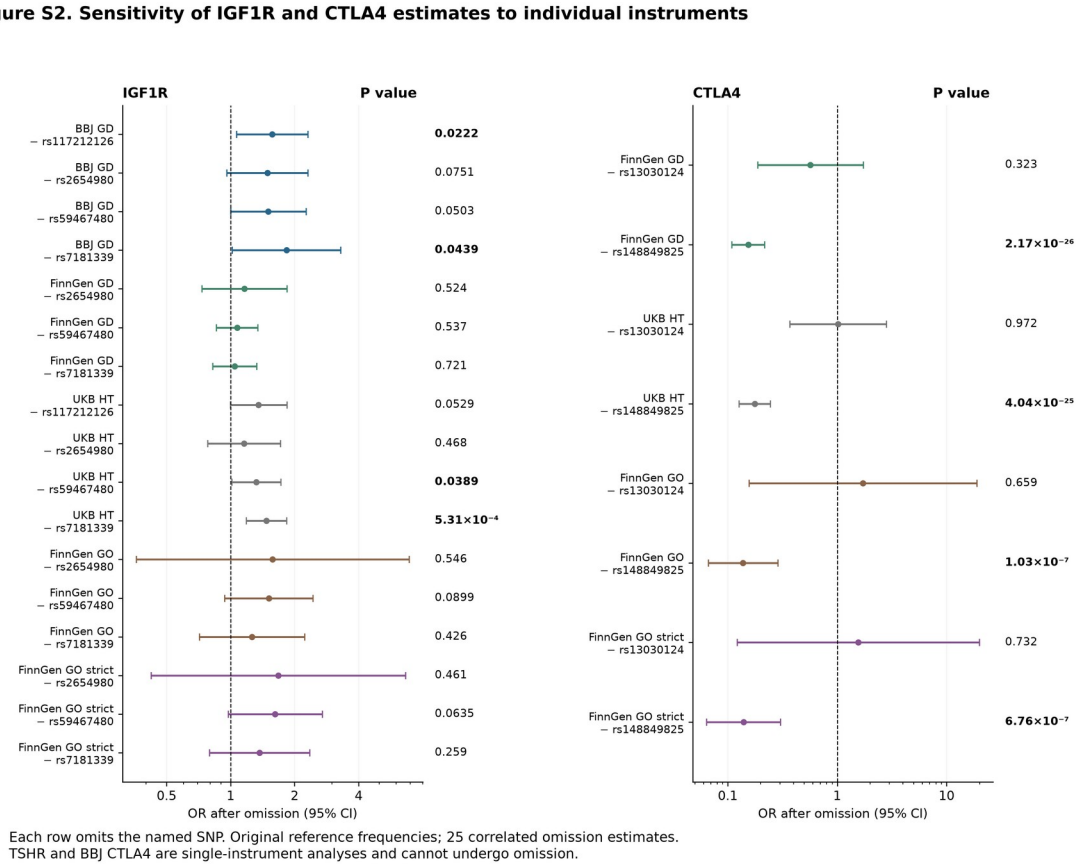


Figure S2. Sensitivity of *IGF1R* and *CTLA4* estimates to individual instruments. Twenty-five reference-frequency SNP-omission estimates for *IGF1R* and *CTLA4* across the five datasets. The named SNP is omitted; Wald ratios apply when one remains and IVW when at least two remain. Horizontal lines show 95% CIs; the vertical reference is OR = 1. Bold *P* values mark nominal $P < 0.05$ only. Original full-set estimates are in Table 2. *TSHR* and BBJ *CTLA4* were ineligible because each has one instrument. These correlated post hoc diagnostics are not independent replication.